

DocuMind course roadmap - 18 chapters, 60 scenes, in the order to build them (22 September 2026)

The working plan for authoring and running the course from v5 (`course-plan-v5-story-2026-09-22.md`, which holds the files, make targets and proofs per scene). Follow it top to bottom. A scene is done when its six files exist, pass the checker and its proof has been rehearsed once on a live lane. A chapter is done when its gate is green.

How to follow it

- **Order.** Phase 0 first. Then chapters 1 to 14 in order (Core, one working system). Then 15 to 18 in any order (Advanced).
- **A scene page has five parts:** Where we are, What we add and why, The code, Prove it, What changed.
- **Six files per scene:** `README.md`, the main page, the practice lab, the live session, the interview Q&A, and a fifth file - a Colab notebook in the Colab chapters, a Cloud Shell runbook in the make-driven chapters (2, 4, 13, 14, 18).
- **The authoring loop, per chapter:** one brief (template, voice, sources, a per-scene outline); one writer per scene, in parallel; the checker (structure, verbatim code, real make targets, clean notebooks); a human read of every main page; one pull request per chapter; the proofs rehearsed on a live lane before the chapter is marked Done.
- **Status:** Not started, Drafted, In review, Done. Chapters 3 and 4 start as Drafted: the five lessons authored on 21 September under v4 numbering (`Module 3/`) hold their content and are re-cut, not rewritten.

Phase 0 - before any authoring

- [] Rehearse `make up` end to end on a throwaway project and record what breaks. The kit's README says the one-shape apply has not run live; every chapter from 2 on is written against its outputs.
- [] Renumber `course-manifest.json` to v5 (18 modules, 60 lessons) and create the folder skeleton, one `README.md` per scene.
- [] Re-cut the authored Module 3 into chapters 3 and 4 (v4 3.1 and 3.3 into chapter 3; v4 3.2, 3.4 and 3.5 into chapter 4).
- [] Rename `deploy/` to `code/` in the tools (`build_learners.py`, `check_contract.py`, `check_auth_wiring.py`, `deploy_index.py`) and re-mirror the kit to the learner repo, so the notebooks' clone matches the code.
- [] Land the four kit fixes found while authoring Module 3: the reindex smoke's tenant filter, the seeder's embedding stamp, the stale comments (attempts, poison ack, page count, cache record), the learner mirror.
- [] Generalise the Module 3 checker to every chapter and run it in the dryrun workflow.
- [] Confirm the two standing choices: 90 minutes per Core scene; Streamlit stays the UI.

Act I - Run it

Chapter 1 - Setup - Run and understand DocuMind (Colab · gate: make dryrun green)

#	Scene	Proof	Fifth file	Status
1.1	Reproduce the local environment and read the master diagram	the Rs 0 lane answers the notice-period question with citations	notebook	Not started
1.2	Prove a first success and diagnose a first failure	ten PASS lines, one FAIL with its reason, green again	notebook	Not started

Chapter 2 - Deployment - Use the cloud lab safely (Cloud Shell · gate: make smoke green, make off floors to zero)

#	Scene	Proof	Fifth file	Status
2.1	Understand the project, identities and resource map	make plan lists every resource; each service's account named	runbook	Not started
2.2	Review and start the prepared deployment	seven services up; the UI behind IAP	runbook	Not started

#	Scene	Proof	Fifth file	Status
2.3	Check readiness, save progress and close the session	smoke green; zero instances after make off; a restored session	runbook	Not started

Act II - Make it know things

Chapter 3 - Ingestion - Turn documents into searchable evidence (Colab · gate: a row with every stamp; vectorsCount moved)

#	Scene	Proof	Fifth file	Status
3.1	Understand source, tenant, page and chunk contracts	a re-wrapped paragraph keeps its hash; a root-level object refused	notebook	Drafted
3.2	Parse documents and compare chunk boundaries	65 identical chunks from both chunkers	notebook	Drafted
3.3	Create and validate compatible embeddings	three 768-number vectors and the cost in paise; a legacy row rejected	notebook	Drafted

#	Scene	Proof	Fifth file	Status
3.4	Write , inspect and verify indexed records	a chunks row with every stamp; vectorsCount moves	notebook	Drafted

Chapter 4 - Lifecycle - Deliver, update and recover documents (Cloud Shell · gate: make smoke-reindex)

#	Scene	Proof	Fifth file	Status
4.1	Follow upload events, retrievals, dead-letter handling and the batch lane	ingest_poison, then the message in the DLQ; a queued claim drained	runbook	Drafted
4.2	Reindex a changed section and measure embedding reuse	reused=281 embedded=2 retired=283; the answer moves	runbook	Drafted
4.3	Publish versions, retire documents and reject stale events	ingest_stale_event; withdrawn after make retire	runbook	Drafted

#	Scene	Proof	Fifth file	Status
4.4	Restore documents and reconcile index differences	<code>ingest_reactivated ... embedded=0;</code> <code>make smoke-reindex green</code>	runbook	Drafted

Act III - Make it answer

Chapter 5 - Retrieval - Retrieve the right evidence (Colab · gate: ablation recorded; `make smoke` refuses a fallback answer)

#	Scene	Proof	Fifth file	Status
5.1	Apply query embeddings and authorized filters	an unknown filter key answered 400; <code>stages.retrieval_backend</code> on the answer	notebook	Not started
5.2	Compare dense and hybrid retrieval	<code>recall@k</code> and MRR per arm on the golden set	notebook	Not started
5.3	Rerank and inspect retrieved candidates	p95 rerank in <code>make usage</code> ; <code>found_by</code> on a citation	notebook	Not started

#	Scene	Proof	Fifth file	Status
5.4	Test fallback with out losing tenant or metadata filters	found_by: firestore with the filter still applied	notebook	Not started

Chapter 6 - Generation - Produce grounded answers and a usable interface (Colab · gate: upload to answer in the UI)

#	Scene	Proof	Fifth file	Status
6.1	Pack evidence within the context budget	the packed set for one question and its tokens	notebook	Not started
6.2	Generate structured answers, citations and refusals	resolved citations; a refusal on an unanswerable question	notebook	Not started
6.3	Stream answers and handle failures	a stream; one forced fallback line with the answer still served	notebook	Not started
6.4	Complete the Streamlit upload-to-answer journey	upload, the version listed, a cited answer streams	notebook	Not started

Chapter 7 - Evaluation - Measure quality and reject regressions (Colab · gate: make eval-live green on a candidate)

#	Scene	Proof	Fifth file	Status
7.1	Build a useful evaluation dataset	a new row accepted by the offline gate	notebook	Not started
7.2	Separate offline checks, live scoring and LLM judgment	make eval green; live numbers; judge scores	notebook	Not started
7.3	Compare one controlled change against a baseline	a pairwise verdict and the rupee delta	notebook	Not started

Act IV - Make it safe and cheap

Chapter 8 - Security - Verify identity, tenant isolation and safety (Colab · gate: isolation 1.00; a blocked prompt)

#	Scene	Proof	Fifth file	Status
8.1	Trace authenticated identity into tenant membership	the identity and tenant on a usage row	notebook	Not started

#	Scene	Proof	Fifth file	Status
8.2	Test valid access, denied access and cross-tenant requests	401 and 403 side by side; isolation 1.00	notebook	Not started
8.3	Exercise DLP, guardrails and audit behavior	the PAN in the DLP tab; prompt_blocked; the upload in the audit tab	notebook	Not started

Chapter 9 - Caching - Cache without serving stale answers (Colab · gate: the second ask costs Rs 0; a miss after reindex)

#	Scene	Proof	Fifth file	Status
9.1	Compare context caching and answer caching	cached_tokens on a row; model_backend=cache on the second ask	notebook	Not started
9.2	Test cache scope, configuration changes and freshness	the miss after make reindex; a hit again after make cache	notebook	Not started

#	Scene	Proof	Fifth file	Status
9.3	Measure latency, avoided calls and false cache hits	the threshold curve and the avoided calls in rupees	notebook	Not started

Act V - Make it act

Chapter 10 - Agents - Build a tool-using agent (Colab · gate: make smoke-chat)

#	Scene	Proof	Fifth file	Status
10.1	Understand tool contracts and the direct agent loop	retrieve in <code>tool_calls</code>	notebook	Not started
10.2	Implement the main Lang Graph workflow	the refuse node fires on a blocked tool	notebook	Not started
10.3	Diagnose tool arguments, access failures and time outs	<code>refusals</code> non-empty; a timed-out tool reported	notebook	Not started

#	Scene	Proof	Fifth file	Status
10.4	Compare the LangChain and ADK adapters	four brains on /health; four cost lines	notebook	Not started

Chapter 11 - Memory - Persist and isolate conversation state (Colab · gate: the restart check)

#	Scene	Proof	Fifth file	Status
11.1	Distinguish agent state, conversation history and knowledge	the memory-checkpointer warning line	notebook	Not started
11.2	Configure and inspect durable conversation storage	the checkpoint tables; one row per thread	notebook	Not started
11.3	Verify restart recovery and session isolation	the conversation continues after a redeploy; another user finds nothing	notebook	Not started

Chapter 12 - Protocols - Connect external agents through MCP and A2A (Colab · gate: make smoke-mcp, make smoke-agent)

#	Scene	Proof	Fifth file	Status
12.1	Expose, discover and invoke MCP tools	tools/list names four; retrieve from a local client	notebook	Not started
12.2	Deploy MCP and verify authorized access	make smoke-mcp green; another tenant refused	notebook	Not started
12.3	Trace the implemented A2A peer and its permissions	the card refused without a token; message/send completes	notebook	Not started

Act VI - Run it for real

Chapter 13 - Operations - Diagnose failures and control spending (Cloud Shell · gate: tenant_daily in INR; zero instances after make off)

#	Scene	Proof	Fifth file	Status
13.1	Debug a wrong answer through the complete pipeline	a wrong answer traced to its cause	runbook	Not started

#	Scene	Proof	Fifth file	Status
13.2	Reconcile usage events, reports and alerts	tenant_daily equals make usage; an alert tripped	runbook	Not started
13.3	Exercise model routing, budgets and shutdown controls	the tier changes at 85 percent; zero instances after make off	runbook	Not started

Chapter 14 - Release - Release, recover and demonstrate the project (Cloud Shell · gate: rollback timed; the rubric scored)

#	Scene	Proof	Fifth file	Status
14.1	Build and deploy using keyless identity	a build and a deploy with no key	runbook	Not started
14.2	Evaluate and gate the exact candidate revision	the candidate judged; its name recorded	runbook	Not started
14.3	Promote, rollback and repair a controlled failure	rollback under two minutes; make smoke-all green	runbook	Not started

#	Scene	Proof	Fifth file	Status
14.4	Complete the independent capstone and operational handover	the five rubric criteria scored	runbook	Not started

Act VII - Extend it (Advanced)

Chapter 15 - Graph - Graph and managed retrieval (Colab · gate: `found_by: rag_engine; status matches the ledger`)

#	Scene	Proof	Fifth file	Status
15.1	Build graph evidence from source documents	node and edge counts	notebook	Not started
15.2	Compare Firestore and Spanner graphs	the CFO question answered without the word CFO	notebook	Not started
15.3	Configure and query managed retrieval mirrors	<code>found_by: rag_engine;</code> a policy skip for an in tenant	notebook	Not started

#	Scene	Proof	Fifth file	Status
15.4	Compare quality and test update/withdrawal freshness	make managed-status matches after a reindex, an undo, a withdrawal	notebook	Not started

Chapter 16 - Multimodal - Multimodal evidence and media (Colab · gate: make smoke-media)

#	Scene	Proof	Fifth file	Status
16.1	Query a video clip and validate media citations	the clip at its second; make smoke-media green	notebook	Not started
16.2	Exercise implemented Studio and voice features	a generated image and its usage row; an answer read aloud	notebook	Not started

Chapter 17 - Tuning - Managed fine-tuning (Colab · gate: the tuned candidate passes make eval-live)

#	Scene	Proof	Fifth file	Status
17.1	Decide whether tuning is justified and prepare data	the frozen manifest and its row count	notebook	Not started

#	Scene	Proof	Fifth file	Status
17.2	Validate sanitized datasets and run managed tuning	the job id; the endpoint path with its location	notebook	Not started
17.3	Compare the tuned candidate with an uncontaminated baseline	the pairwise verdict and the rupee delta	notebook	Not started

Chapter 18 - Serving - Model gateway and self-hosted inference (Cloud Shell · gate: make smoke-gateway, make smoke-slm; zero GPU instances)

#	Scene	Proof	Fifth file	Status
18.1	Trace and authorize gateway routes	a PAN re-routed; the cost header	runbook	Not started
18.2	Serve a supplied or stock model using Ollama	the cold start timed; a gated answer from the small model	runbook	Not started

#	Scene	Proof	Fifth file	Status
18.3	Inspect the vLLM service and the GKE alternative	the manifest read; the duty-cycle sum	runbook	Not started
18.4	Compare actual back ends and verify shutdown behavior	the comparison table; zero GPU instances	runbook	Not started

Authoring batches - the order of work

Batch	Chapters	Scenes	Needs first	Ends with
A	1, 2	5	the Phase 0 rehearsal	pull requests for chapters 1 and 2
B	3, 4	8	the Module 3 re-cut	pull requests for 3 and 4
C	5, 6, 7	11	a live lane with the corpus loaded	pull requests for 5, 6 and 7
D	8, 9	6	chapter 7's gate in use	pull requests for 8 and 9
E	10, 11, 12	10	the chat service and MCP deployed	pull requests for 10, 11 and 12
F	13, 14	7	the CD workflow installed	pull requests for 13 and 14
G	15, 16, 17, 18	13	managed stores enabled (15); GPU quota (18)	pull requests for 15 to 18

Sixty scenes. At one writer per scene and one batch a week, seven weeks of authoring, plus review and rehearsal.

What done means

- **Scene:** six files; the checker green (5 quizzes, 3 exercise cards, the titles and the template, every code excerpt verbatim, every make target real, the notebook clean); every proof rehearsed once on a live lane; read by a second person.
- **Chapter:** the gate green on the live lane; one pull request merged; the learner repo re-mirrored when a notebook changed.